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ABSTRACT

Multi-agent real-time adversarial control remains challenging because reward signals mix team-level and agent-specific semantics, the action space is structured and constrained, and decision making must be performed under temporal partial observability. To address these issues, this paper proposes a unified reinforcement learning framework named HFO-SERPPO. The framework first introduces a shared-and-exclusive reward formulation and develops SERPPO on top of PPO to improve the alignment between agent behavior and critic targets. It then incorporates pseudo-termination-aware value estimation, structured multi-discrete policy parameterization, and legality-aware action masking to improve optimization stability and policy learning in constrained action spaces. Finally, it extends the framework in the temporal dimension by combining a future-observing critic with a history-aware recurrent actor, yielding the full HFO-SERPPO architecture. Experiments on a RoboMaster-style multi-robot adversarial simulator show that the proposed framework consistently outperforms behavior-tree and MAPPO baselines, while also demonstrating favorable robustness under opponent, team-size, and environment changes. These results indicate that jointly modeling reward semantics, structured decision spaces, and asymmetric temporal information is effective for multi-agent adversarial control under partial observability.

1. Introduction

Recent advances in deep reinforcement learning (DRL) have substantially improved sequential decision making in complex control problems, including robotic control, game playing, and multi-agent coordination [1, 2, 3, 4]. Among these directions, multi-agent real-time adversarial control remains particularly challenging because each agent must make decentralized decisions under strong interactions, dynamically changing opponent behaviors, and incomplete observations. Compared with standard single-agent control tasks, such scenarios require agents not only to learn effective local behaviors, but also to coordinate with teammates and adapt continuously to adversarial policies. These characteristics make multi-agent adversarial control an important yet difficult testbed for studying the practical limits of DRL in interactive robotic decision making [5, 6, 7].

This difficulty becomes more pronounced in robotic adversarial environments with local observations, obstacle constraints, and structured control commands. In such settings, the learning problem is shaped by at least three coupled challenges. First, the reward signal is semantically heterogeneous. Some events, such as victory, defeat, or key tactical outcomes, are naturally shared across the team, whereas many dense shaping signals, such as collision penalties, aiming quality, local maneuver quality, or branch-specific operation penalties, are intrinsically agent-specific. When these heterogeneous signals are merged into a purely shared reward, the value target of each agent becomes less aligned with its own behavior, which can weaken credit assignment and introduce unnecessary ambiguity into critic learning. Second, the decision space is structured rather than atomic. A robot usually needs to determine translational motion, rotational control, target engagement, and attack-related actions jointly, so naively flattening all action combinations into a single discrete output can lead to poor parameterization, inefficient exploration, and unstable optimization. Third, the task is temporally partially observable. A single current observation is often insufficient for inferring short-term motion trends, opponent intent, or evolving local interactions, while the critic is simultaneously affected by the uncertainty induced by the near-future actions of teammates and opponents [8, 9, 10, 11, 12].

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Existing multi-agent reinforcement learning methods address parts of these issues, but important gaps remain. PPO-based methods have become strong practical baselines because of their simplicity, scalability, and empirical effectiveness [2, 13]. Value-decomposition and centralized-critic methods improve coordination in many cooperative settings [14, 15, 16], while recurrent policies and action masking have shown clear benefits under partial observability and constrained action spaces [17, 10, 18]. However, these techniques are often introduced separately. In multi-agent robotic adversarial control, reward semantics, action structure, optimization stability, and temporal reasoning are tightly coupled. Treating them as isolated implementation choices is often insufficient to provide a well-aligned and robust learning framework for real-time adversarial interaction.

To address these issues, this paper proposes a unified reinforcement learning framework for multi-agent real-time adversarial control. We first reformulate the task with a shared-and-exclusive reward structure and build a corresponding *Shared-and-Exclusive Reward PPO* (SERPPO) framework. This formulation preserves team-level cooperation while maintaining clearer agent-level accountability in critic targets, thereby improving the semantic consistency between the reward source and the learning objective of each agent. On top of this formulation, we further improve optimization stability and policy parameterization through pseudo-termination-aware value estimation, structured multi-discrete action modeling, and legality-aware action masking. These components are introduced not as isolated engineering tricks, but as mutually related design choices for stabilizing PPO-style learning under structured, constrained, and partially truncated adversarial rollouts.

The framework is further extended in the temporal dimension. Specifically, we introduce an asymmetric temporal design that combines a future-observing critic with a history-aware recurrent actor, yielding the final *History-and-Future-Observing SERPPO* (HFO-SERPPO) architecture. The motivation is that the actor and critic face different types of uncertainty under partial observability. During execution, the actor must remain causal and therefore needs to infer latent context from historical observations. During training, the critic may benefit from richer temporal cues that help reduce ambiguity in value regression under rapidly evolving multi-agent interactions. By using historical information on the actor side and short-horizon future interaction cues on the critic side, the proposed framework improves temporal reasoning without violating decentralized execution requirements.

The resulting method is evaluated on a RoboMaster-style multi-robot adversarial simulator with local observations and obstacle constraints. This setting is representative of robotic competitive-cooperative interaction, where agents must jointly handle motion control, obstacle avoidance, target selection, and tactical engagement. The experimental results show that the proposed framework consistently outperforms behavior-tree and MAPPO baselines, and remains robust under changes in opponent policy, team size, and environment layout.

The main contributions of this paper are summarized as follows:

- We propose a shared-and-exclusive reward formulation for multi-agent adversarial control and develop the SERPPO framework on top of PPO. This design improves the semantic alignment between reward signals and agent-level learning targets, leading to more effective credit assignment.
- We introduce a set of optimization-oriented enhancements, including pseudo-termination-aware value estimation, structured multi-discrete policy parameterization, and legality-aware action masking. These components improve critic fitting quality, reduce ineffective exploration, and make structured action spaces more tractable for PPO-based learning.
- We develop an asymmetric temporal extension for partial observability by combining a future-observing critic with a history-aware recurrent actor, resulting in HFO-SERPPO. This design improves learning efficiency and decision quality in temporally evolving adversarial interactions.

The remainder of this paper is organized as follows. Section 2 reviews related studies. Section 3 introduces the necessary preliminaries. Section 4 presents the proposed framework in detail. Section 5 reports the experimental results and analysis. Finally, Section 6 concludes the paper and discusses future work.

2. Related work

2.1. Multi-agent reinforcement learning for adversarial decision making

Multi-agent adversarial control is commonly addressed by either manually designed controllers or learning-based policies. In robotic and game-like control tasks, behavior trees remain a practical baseline because of their interpretability, modularity, and reactive execution characteristics [19]. They are effective when tactical rules are well understood and controllability is emphasized, but their adaptability is limited once opponent behavior, interaction patterns, or environment configurations change substantially.

Learning-based methods attempt to optimize policies directly from interaction. In cooperative and mixed cooperative-competitive settings, value-factorization methods such as VDN [14], QMIX [15], QTRAN [20], QPLEX [21], MAVEN [22], and ROMA [23] improve coordination by decomposing or regularizing the value function. Another influential line is actor-critic-based MARL, represented by MADDPG [16] and MAPPO [13], which are often more flexible in continuous or structured control settings. Recent work has also shown that practical implementation details can significantly affect the empirical behavior of cooperative MARL algorithms [24].

Reinforcement learning has also been explored in robotic adversarial or competitive control, including multi-UAV confrontation, competitive maneuvering, and multi-robot decision making [6, 25, 7]. These studies demonstrate the promise of learning-based control in dynamic adversarial environments. However, many existing approaches either rely on standard MARL backbones or focus on task-specific decomposition, without explicitly addressing the joint effect of mixed reward semantics, structured action spaces, and temporal partial observability that characterizes real-time robotic adversarial control.

2.2. Reward design, structured actions, and stable policy optimization

A second line of related work focuses on improving policy optimization through better credit assignment, reward design, and action-space modeling. In multi-agent policy-gradient methods, credit assignment is a longstanding challenge because global performance depends on joint behavior while each agent observes and controls only part of the system. Counterfactual actor-critic methods such as COMA alleviate this issue through agent-level advantage estimation under a centralized critic [8]. Related studies have also explored reward decomposition or synchronized reward mechanisms for multi-agent learning [26, 27]. These results suggest that the semantic structure of reward signals strongly influences what coordination patterns can be learned.

Another closely related issue is structured action modeling. In many robotic and game-like tasks, the action is a combination of several semantically different control branches rather than a single atomic decision. Large-scale game RL systems therefore often adopt structured or factorized action representations instead of monolithic action-combination modeling [11, 12]. A similar idea appears in action-branching architectures, which improve tractability by decomposing high-dimensional decision spaces into multiple sub-decisions [28]. In addition, invalid action masking has proved effective for excluding temporarily infeasible actions and reducing ineffective exploration in constrained policy optimization [10, 18].

Stable value estimation is also closely tied to rollout handling and bootstrap design. In on-policy actor-critic methods, rollout truncation and true environmental termination have different learning semantics, but they are not always treated distinctly in practice. Prior work has discussed how time limits and artificial truncation can bias value targets if they are handled as genuine terminal transitions. [29] This issue is especially relevant to PPO-style training, where critic fitting quality directly affects advantage estimation and policy updates.

Overall, prior studies have shown the importance of credit assignment, reward structure, action factorization, and legality constraints. However, these ideas are often introduced separately. Comparatively less attention has been paid to a unified formulation in which mixed shared-exclusive rewards, structured multi-discrete decisions, masking-based feasibility constraints, and truncation-aware value estimation are designed jointly for adversarial multi-agent control.

2.3. Temporal modeling under partial observability

Partial observability is another central challenge in multi-agent control. When each agent receives only local observations, a single time step is often insufficient for inferring recent motion tendencies, opponent intent, or cooperative context. Recurrent reinforcement learning therefore provides a natural way to aggregate temporally dispersed evidence. Early work such as DRQN demonstrated the effectiveness of recurrence in partially observable sequential decision problems

[17]. Later studies extended this direction to multi-agent settings through recurrent or belief-based architectures, including attention-based recurrent MARL and common-belief models [9, 30]. These studies show that temporal aggregation is an effective mechanism for mitigating observation incompleteness.

A related stream improves partial observability and coordination through communication or richer training-time information. Learned communication mechanisms [31, 32] enable agents to exchange task-relevant information, while other work studies asynchronous training or recurrent policy optimization under temporally dependent decision processes [33, 34]. In actor–critic methods, asymmetric information usage has also been explored, where the critic is allowed to access richer training-time signals than the actor. [35] This line is particularly relevant to settings in which the actor must remain causal at execution time, while the critic can exploit privileged information to reduce ambiguity during learning.

Existing work therefore provides useful tools for temporal reasoning under partial observability, but most studies focus on either recurrent policy design or centralized training in a broad sense. Comparatively less attention has been paid to asymmetric temporal modeling for adversarial multi-agent PPO, where historical information is aggregated on the actor side while short-horizon future interaction cues are injected only into the critic during training.

Taken together, prior studies have established strong foundations in adversarial MARL, optimization-oriented policy design, and temporal modeling under partial observability. Nevertheless, these directions are still usually developed in parallel. In contrast, the present work combines a shared-and-exclusive reward formulation, truncation-aware and structure-aware policy optimization, and an asymmetric history–future temporal design within a unified PPO-based framework for multi-agent real-time adversarial control.

3. Preliminaries

This section introduces the notation and basic optimization background required for the proposed method. For notational convenience, we first adopt the standard reinforcement-learning description and then clarify the multi-agent partially observable setting considered in this paper. Since the subsequent method is built on a PPO-style actor–critic framework, we also briefly review the corresponding policy objective and advantage estimation.

We consider a multi-agent sequential decision problem with N agents. At each time step t , agent i receives a local observation $o_{i,t}$ and selects an action $a_{i,t}$ according to a policy π_i . The joint action of all agents is denoted by $\mathbf{a}_t = (a_{1,t}, \dots, a_{N,t})$. For notation, a standard reinforcement-learning task can be described by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{P}$ denotes the transition dynamics, $\mathcal{R}$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. The learning objective is to optimize the policy so as to maximize the expected discounted return.

In the actual setting studied in this paper, however, the problem is more naturally viewed as a decentralized partially observable multi-agent control problem rather than a fully observable centralized one. Each agent makes decisions from its own local observation, while richer training-side information may be available for value estimation. To keep the notation clear in the following sections, we therefore distinguish two forms of input. The actor of agent i receives the local observation $o_{i,t}$, whereas the critic uses an agent-centric training state $s_{i,t}$ for value estimation. Here, $s_{i,t}$ does not necessarily denote the full global environment state; rather, it denotes the training-time state representation used to evaluate the return relevant to agent i .

For agent i , the discounted return from time step t is defined as

$$G_{i,t} = \sum_{j=t}^T \gamma^{j-t} r_{i,j}, \quad (1)$$

where $r_{i,j}$ is the reward received by agent i at time step j , and T is the terminal time step of the episode. Based on this return, the state-value function is written as

$$V_i^\pi(s_{i,t}) = \mathbb{E}_\pi [G_{i,t} \mid s_{i,t}], \quad (2)$$

and the action-value function is

$$Q_i^\pi(s_{i,t}, a_{i,t}) = \mathbb{E}_\pi [G_{i,t} \mid s_{i,t}, a_{i,t}]. \quad (3)$$

These definitions provide the basic value-estimation notation used throughout the rest of the paper.

The proposed framework is built on an actor–critic formulation with PPO-style policy optimization [2, 13]. In this framework, the actor parameterizes the policy, while the critic estimates the value function and provides learning signals for policy improvement. Let θ denote the actor parameters and θ_{old} the behavior-policy parameters used for sample collection. PPO updates the policy through a clipped surrogate objective:

$$\mathcal{L}_{\text{clip}} = \mathbb{E} \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (4)$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t | o_t)}{\pi_{\theta_{\text{old}}}(a_t | o_t)}, \quad (5)$$

and $\hat{A}_t$ is the estimated advantage. The clipping mechanism constrains the policy update so that the new policy does not deviate excessively from the sampling policy, which is one of the main reasons for the practical stability of PPO in on-policy learning.

Advantage estimation plays a central role in PPO-based training because it connects value estimation with policy optimization. A widely used choice is generalized advantage estimation (GAE) [36], which recursively accumulates temporal-difference information. For agent i , the temporal-difference error at time step t is

$$\delta_{i,t} = r_{i,t} + \gamma V_i(s_{i,t+1}) - V_i(s_{i,t}), \quad (6)$$

and the corresponding advantage estimate is

$$\hat{A}_{i,t} = \delta_{i,t} + \gamma \lambda \hat{A}_{i,t+1}, \quad (7)$$

where $\lambda \in [0, 1]$ is the trace-decay parameter. This estimator provides a practical bias–variance trade-off and is therefore widely adopted in PPO-style actor–critic methods.

The above notation is sufficient for the method development in the next section. In summary, the actor produces decentralized actions from local observations, the critic estimates agent-related returns from training states, and policy learning follows a PPO-style clipped objective with GAE-based advantage estimation. On top of this common actor–critic basis, the next section introduces the reward formulation, optimization-oriented modifications, and temporal extensions specific to the proposed framework.

4. Method

This section presents the proposed framework for multi-agent real-time adversarial control. The method is developed in three progressively connected stages. We first introduce a shared-and-exclusive reward formulation and build the corresponding SERPPO framework. We then improve optimization stability and policy parameterization through truncation-aware value estimation, structured multi-discrete action modeling, and legality-aware action masking. Finally, we extend the framework in the temporal dimension with a future-observing critic and a history-aware recurrent actor, resulting in the full HFO-SERPPO architecture. A distributed rollout strategy is further used to improve training throughput, and the complete training pipeline is summarized at the end of this section. The overall data flow and module interactions of the proposed framework are illustrated in Fig. 1.

4.1. Shared-and-exclusive reward formulation and SERPPO

PPO-based actor–critic methods provide a strong and practical backbone for multi-agent reinforcement learning [2, 13]. However, in multi-agent adversarial robotic control, an important difficulty is that the reward signal is semantically heterogeneous. In addition to team-level outcomes such as victory, defeat, or decisive tactical events, many dense shaping terms are inherently agent-specific, including local collision penalties, aiming quality, and fine-grained maneuver quality. If all these components are directly aggregated into a purely shared reward, the value target of each agent becomes only partially attributable to its own behavior, which weakens credit assignment and makes critic learning more ambiguous.

To address this issue, we model the task with a shared-and-exclusive reward structure. For agent i , the reward at time step t is decomposed as

$$r_{i,t} = r_t^{\text{sh}} + r_{i,t}^{\text{ex}}, \quad (8)$$

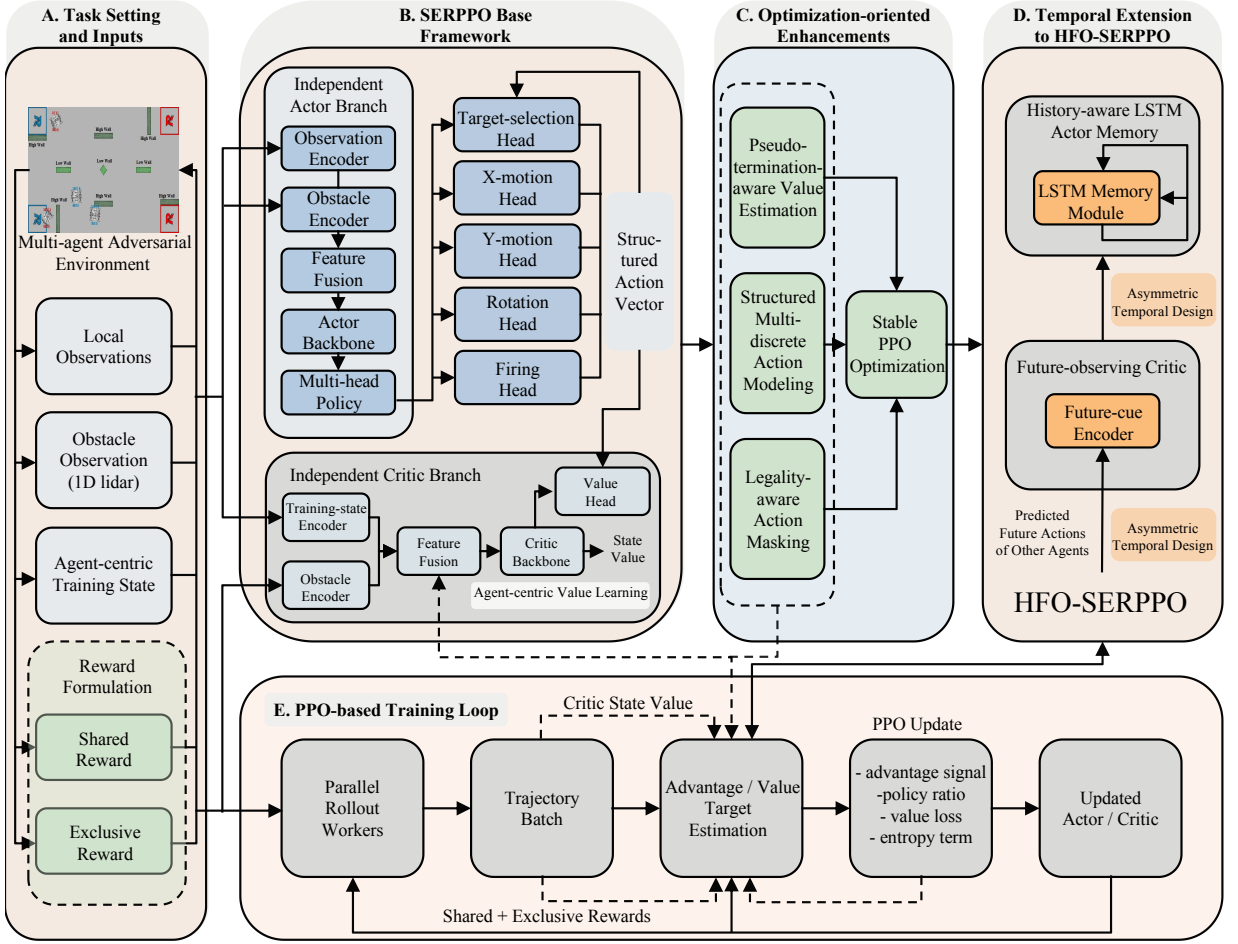


Fig. 1: Overall framework of HFO-SERPPO. The figure summarizes the task inputs, the SERPPO base framework, the optimization-oriented enhancements, the temporal extensions, and the PPO-based training loop in a unified view.

where r_t^{sh} is the team-shared reward and $r_{i,t}^{\text{ex}}$ is the exclusive reward associated only with agent i . Accordingly, the return of agent i is

$$G_{i,t} = \sum_{j=t}^T \gamma^{j-t} r_{i,j}. \quad (9)$$

This formulation preserves two complementary aspects of the task: cooperation through the shared component and agent-level accountability through the exclusive component.

Based on this formulation, we develop *Shared-and-Exclusive Reward PPO* (SERPPO). The actor follows the PPO-Clip update rule. For agent i , the importance ratio is

$$\rho_{i,t}(\theta) = \frac{\pi_{\theta}(a_{i,t} | o_{i,t})}{\pi_{\theta_{\text{old}}}(a_{i,t} | o_{i,t})}, \quad (10)$$

and the clipped surrogate objective is

$$\mathcal{L}_{\text{clip}} = \mathbb{E}_{i,t} \left[\min \left(\rho_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_{i,t} \right) \right]. \quad (11)$$

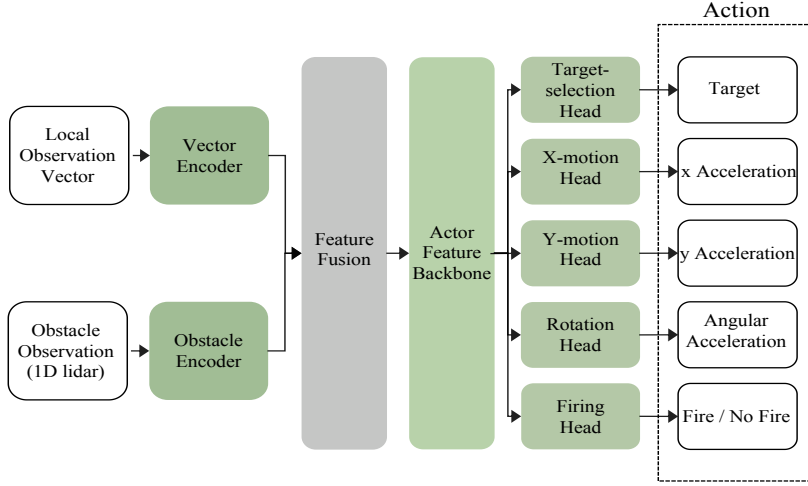


Fig. 2: Independent actor architecture of SERPPO with factorized multi-head policy outputs.

The key difference from a conventional shared-reward multi-agent PPO framework lies in the critic target. Instead of estimating a purely shared team return, the critic of agent i is trained to fit an agent-centric return:

$$V_i(s_{i,t}) \approx \mathbb{E}[G_{i,t} \mid s_{i,t}], \quad (12)$$

where $s_{i,t}$ denotes the agent-centric training state introduced in Section 3. Since the return in Eq. (9) contains both the team-shared reward and the exclusive reward of the corresponding agent, the value target remains better aligned with the effective reward source relevant to agent i . In this sense, SERPPO changes not only the optimization target but also the semantic structure of credit assignment in multi-agent adversarial learning.

The overall objective of SERPPO is written as

$$\mathcal{L}_{\text{SERPPO}} = \mathcal{L}_{\text{clip}} - c_v \mathcal{L}_{\text{vf}} + c_e \mathcal{L}_{\text{ent}}, \quad (13)$$

where $\mathcal{L}_{\text{vf}}$ is the value regression loss and $\mathcal{L}_{\text{ent}}$ is the entropy regularization term. The independent actor and critic architectures adopted in SERPPO are shown in Figs. 2 and 3, respectively. The actor uses local observations and obstacle observations to produce a factorized multi-head policy, whereas the critic uses agent-centric training states together with obstacle observations for value estimation.

4.2. Formula-level enhancements for stable optimization

4.2.1. Pseudo-termination-aware value estimation

In on-policy training, a collected rollout segment may end for two different reasons: the environment may genuinely terminate, or the rollout may be truncated at a fixed horizon for collection efficiency. These two cases have different learning semantics. Assigning zero bootstrap value to both is correct only for true termination. For truncated segments, the next state may still contain substantial future value, and treating it as terminal can bias critic targets and, consequently, GAE-based advantage estimation.

To distinguish these two cases, each stored transition is augmented with a pseudo-termination indicator:

$$(o_{i,t}, s_{i,t}, a_{i,t}, r_{i,t}, o_{i,t+1}, d_t, p_t), \quad (14)$$

where d_t indicates the end of the sampled segment and p_t indicates whether that endpoint corresponds to a true environmental termination.

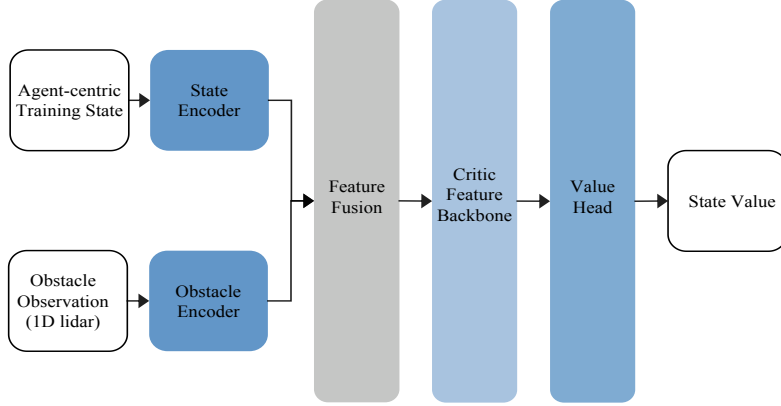


Fig. 3: Independent critic architecture of SERPPO for agent-centric value estimation.

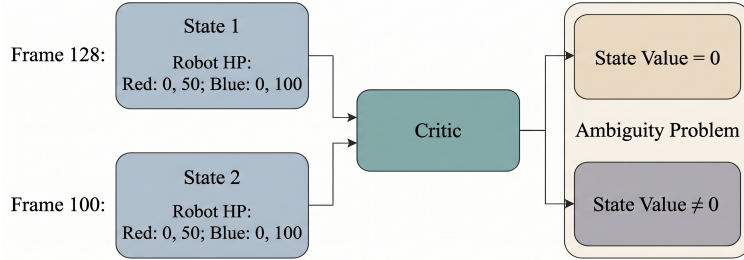


Fig. 4: Illustration of critic ambiguity caused by conflating true termination with rollout truncation.

This distinction is closely related to prior discussions on time limits and artificial truncation in reinforcement learning, where incorrect terminal handling can bias bootstrap targets [29]. In our setting, the last-step advantage is therefore computed as

$$\hat{A}_{i,T'} = \begin{cases} r_{i,T'} - V_i(s_{i,T'}), & p_{T'} = \text{True}, \\ r_{i,T'} + \gamma V_i(s_{i,T'+1}) - V_i(s_{i,T'}), & p_{T'} = \text{False}, \end{cases} \quad (15)$$

while preceding steps follow the standard recursive GAE form

$$\hat{A}_{i,t} = \delta_{i,t} + \gamma \lambda \hat{A}_{i,t+1}, \quad (16)$$

with

$$\delta_{i,t} = r_{i,t} + \gamma V_i(s_{i,t+1}) - V_i(s_{i,t}). \quad (17)$$

The practical intuition is illustrated in Fig. 4. Two endpoint observations may appear similarly “terminal-like” at the end of a sampled rollout, while having fundamentally different value semantics: one may correspond to a genuinely finished confrontation, whereas the other may simply be a sampling cutoff. Distinguishing these cases reduces target ambiguity for the critic and improves the consistency of value estimation.

4.2.2. Structured multi-discrete policy and action masking

The control action in the considered task is structured rather than atomic. Each robot must jointly determine translational movement, rotational control, target selection, and attack-related behaviors. Flattening all action combinations into a single discrete output would unnecessarily enlarge the policy space and obscure the semantic role of each control branch. Similar motivations appear in large-scale game reinforcement learning and action-branching architectures, where factorized decision modeling is preferred over monolithic action-combination representations in high-dimensional decision spaces [11, 12, 28].

Accordingly, we model the action as a multi-discrete vector:

$$a_{i,t} = \left(a_{i,t}^x, a_{i,t}^y, a_{i,t}^\omega, a_{i,t}^{\text{shoot}}, a_{i,t}^{\text{target}} \right), \quad (18)$$

and parameterize the policy through a factorized multi-head distribution:

$$\pi_\theta(a_{i,t} | o_{i,t}) \approx \prod_{m=1}^M \pi_\theta^{(m)}(a_{i,t}^{(m)} | o_{i,t}). \quad (19)$$

The corresponding log-probability becomes

$$\log \pi_\theta(a_{i,t} | o_{i,t}) = \sum_{m=1}^M \log \pi_\theta^{(m)}(a_{i,t}^{(m)} | o_{i,t}), \quad (20)$$

which can be used directly in the PPO ratio computation in Eq. (10).

This factorization does not assume that different control branches are physically independent in the environment. Rather, it provides a more suitable policy parameterization for structured decisions by combining a shared feature backbone with multiple semantically meaningful output heads. This design improves tractability while preserving interpretability and branch-level control structure.

To further improve learning efficiency, legality-aware action masking is introduced for branches with explicit feasibility constraints, especially target selection and shooting. Invalid action masking has been shown to reduce ineffective exploration in constrained decision problems, particularly in PPO-style optimization [10, 18]. Let $M_{i,t}^{(m)}$ denote the availability mask of action branch m . The masked logits are written as

$$\tilde{z}_{i,t}^{(m)} = z_{i,t}^{(m)} + (1 - M_{i,t}^{(m)})(-\infty), \quad (21)$$

followed by branch-wise softmax normalization. In the final framework, masking is applied conservatively so that probability mass is prevented from being allocated to temporarily infeasible actions, while the structured multi-discrete parameterization is kept unchanged. Therefore, action factorization and legality-aware masking jointly form a coherent structure-aware policy design for adversarial robotic control.

4.3. History-future temporal extension: HFO-SERPPO

Although SERPPO improves reward semantics and policy parameterization, the task remains temporally partially observable. A single current observation is often insufficient for inferring recent motion tendencies, opponent intent, or short-term cooperative patterns. Meanwhile, value regression is affected by uncertainty induced by the near-future actions of teammates and opponents. To address these two distinct issues, we introduce an asymmetric temporal extension that combines a future-observing critic with a history-aware recurrent actor. Removing these temporal components yields the base SERPPO framework, while including them results in the full HFO-SERPPO model.

4.3.1. Future-observing critic

During training, the critic is augmented with short-horizon future actions of other agents:

$$V_{i,t}^{\text{FO}} = V_i \left(s_{i,t}, \{a_{t+j}^{-i}\}_{j=0}^k \right), \quad (22)$$

where a_{t+j}^{-i} denotes the joint action of all agents other than agent i at time $t + j$, and k is the future horizon. The motivation is that a substantial part of the uncertainty in value regression comes from imminent interactions that are not

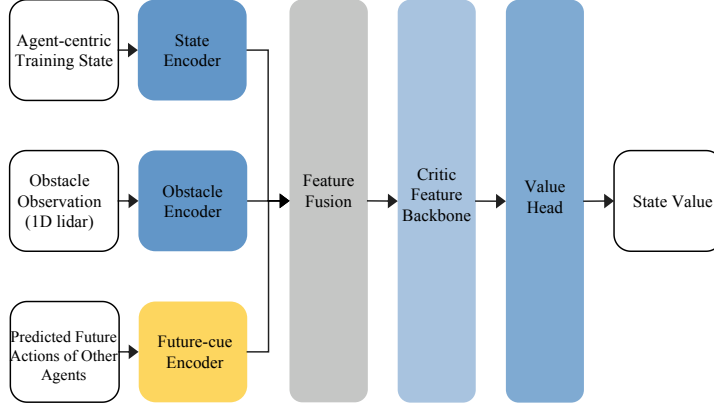


Fig. 5: Network architecture of the future-observing critic. The critic incorporates predicted future actions of other agents through a future-cue encoder for temporally enriched value estimation during training.

fully inferable from the current state alone. Conditioning the critic on short-horizon future actions makes the regression target less ambiguous, especially in adversarial multi-agent scenarios where local outcomes are highly sensitive to nearby short-term interactions.

This design is consistent with the general idea of asymmetric information usage in actor–critic learning, where the critic may exploit richer training-time information than the actor [35, 16, 13]. Importantly, the future interaction cue is used only during training. It is never exposed to the actor during execution and therefore does not violate causality or decentralized deployment requirements. The detailed network architecture of the future-observing critic is shown in Fig. 5, where predicted future actions of other agents are encoded as additional critic-side temporal cues.

4.3.2. History-aware LSTM actor

In contrast to the critic, the actor must remain causal at execution time. We therefore enhance the actor with historical context through an LSTM-based recurrent policy rather than by exposing it to future information. Recurrent policies have been widely used to alleviate partial observability by aggregating temporally dispersed evidence rather than relying solely on the current observation [17, 9, 34]. Let x_t denote the encoded current observation. The recurrent hidden state is updated as

$$h_t = f(x_t, h_{t-1}; w_h), \quad (23)$$

and the policy is generated by

$$\pi_\theta(a_{i,t} \mid o_{i,t}, h_{t-1}) = \Pi(h_t), \quad (24)$$

where $\Pi(\cdot)$ denotes the multi-head action predictor. This design allows the actor to accumulate short-term temporal evidence that is useful for recognizing motion trends, aiming intent, and evolving local interactions from observation sequences. The detailed network architecture of the history-aware LSTM actor is shown in Fig. 6, where the recurrent memory module is inserted between the actor backbone and the multi-head policy outputs.

The future-observing critic and the history-aware actor play complementary roles. The former reduces ambiguity in value learning during training, whereas the latter improves policy inference under execution-time partial observability. Their combination yields the final *History-and-Future-Observing SERPPO* (HFO-SERPPO) framework.

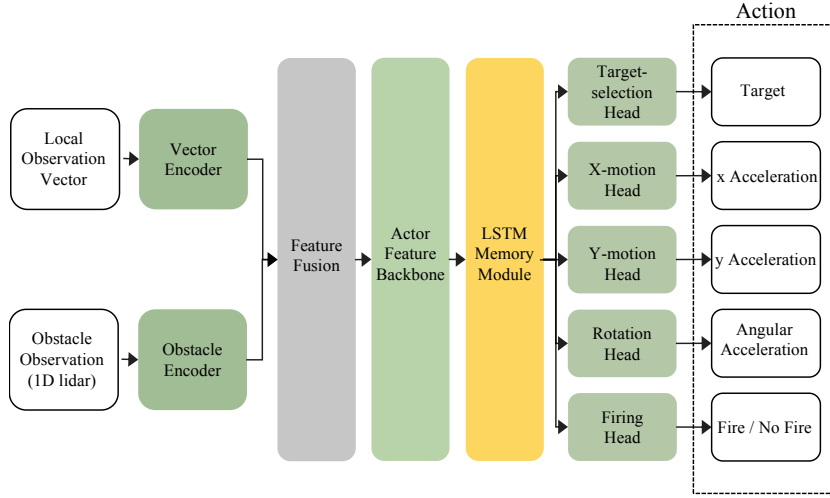


Fig. 6: Network architecture of the history-aware LSTM actor. The recurrent memory module enhances sequential policy inference before the factorized multi-head action outputs.

4.4. Distributed rollout collection

The above components define the learning objective and temporal design of the framework, while training throughput is further improved through distributed rollout collection. Since environment interaction is a major bottleneck in multi-agent adversarial reinforcement learning, multiple simulator instances are executed in parallel to collect trajectory fragments simultaneously. These fragments are then aggregated into a unified on-policy batch for policy optimization. Such parallel data collection is a common practical strategy in large-scale or partially observable reinforcement learning systems [12, 33].

Let $\mathcal{D}^{(w)}$ denote the trajectory set generated by worker w . The global training batch is

$$\mathcal{D} = \bigcup_{w=1}^W \mathcal{D}^{(w)}, \quad (25)$$

where W is the number of rollout workers. This mechanism does not change the optimization objective itself; it improves sample collection throughput and makes the full framework practically trainable at a larger scale.

4.5. Overall training procedure

The final training pipeline integrates the shared-and-exclusive reward formulation, pseudo-termination-aware advantage estimation, structured multi-discrete policy learning, legality-aware action masking, recurrent actor modeling, future-observing critic design, and distributed rollout collection. SERPPO can be viewed as the base version of the framework, while HFO-SERPPO is obtained by further introducing the history-aware actor and the future-observing critic. The overall data flow and module interactions of the proposed framework are summarized in Fig. 1, while the corresponding training procedure of the full model is given in Algorithm 1.

Overall, the proposed method can be understood as a unified PPO-based framework for adversarial multi-agent control. SERPPO provides the shared-and-exclusive reward formulation together with optimization-oriented refinements, whereas HFO-SERPPO further extends this base framework through asymmetric temporal modeling of the actor and critic. The empirical behavior of these components is examined in the next section.

Algorithm 1 HFO-SERPPO for multi-agent real-time adversarial control

Input: actor parameters θ , critic parameters ϕ , number of rollout workers W , rollout horizon H , future horizon k , recurrent chunk length l

Output: optimized actor parameters θ and critic parameters ϕ

```
1: for each training iteration do
2:   Broadcast current parameters  $(\theta, \phi)$  to all rollout workers
3:   for each worker  $w = 1, \dots, W$  in parallel do
4:     Reset environment states and recurrent hidden states
5:     for  $t = 1$  to  $H$  do
6:       Obtain local observation  $o_{i,t}$  and training state  $s_{i,t}$  for each agent  $i$ 
7:       Encode  $o_{i,t}$  and update recurrent hidden state  $h_{i,t}$ 
8:       Compute multi-head action logits for each branch
9:       Apply legality-aware action masks to constrained branches
10:      Sample factorized action  $a_{i,t}$  and execute the joint action
11:      Receive  $o_{i,t+1}$ ,  $r_{i,t}$ , segment-end flag  $d_t$ , and pseudo-termination flag  $p_t$ 
12:      Store  $(o_{i,t}, s_{i,t}, a_{i,t}, r_{i,t}, o_{i,t+1}, d_t, p_t)$ 
13:    end for
14:    Construct future-action sequences  $\{a_{t+j}^{-i}\}_{j=0}^k$  for critic training, with zero padding when needed
15:  end for
16:  Aggregate all worker trajectories into a unified on-policy batch  $\mathcal{D}$ 
17:  Compute pseudo-termination-aware GAE using Eqs. (15)–(17)
18:  Split recurrent trajectories into chunks of length  $l$ 
19:  for each PPO epoch do
20:    for each mini-batch of chunks do
21:      Compute the factorized policy ratio using Eqs. (10) and (20)
22:      Compute the clipped policy loss  $\mathcal{L}_{\text{clip}}$ 
23:      Compute the value loss using the agent-centric critic targets
24:      Compute entropy regularization
25:      Update  $(\theta, \phi)$  using Eq. (13)
26:    end for
27:  end for
28: end for
```

5. Experiments and analysis

5.1. Experimental settings

We evaluate the proposed framework on a 2D RoboMaster-style multi-agent adversarial simulator, where two allied robots confront two enemy robots under local observation, obstacle constraints, and dynamically changing opponent behaviors. This setting is representative of real-time robotic adversarial control because effective decision making requires coordinated motion, target engagement, and obstacle-aware interaction under decentralized perception [37, 38]. Unless otherwise stated, experiments are conducted on the default 2-versus-2 setting with the complete obstacle layout, and each reported evaluation statistic is computed over 100 combat episodes. The learned policy is trained and tested under local observations, while richer training-side information is used only where required by the critic design in the proposed framework.

The compared methods include both external baselines and internal variants of the proposed framework. As external baselines, we consider an aggressive behavior-tree policy, a defensive behavior-tree policy, and MAPPO. These baselines are chosen because behavior-tree policies remain practically relevant scripted controllers in robotic or game-like control, whereas MAPPO is a strong PPO-based MARL baseline with competitive empirical performance [19, 13]. As internal variants, we consider SERPPO-noPD, which removes pseudo-termination-aware value estimation; SERPPO-1DActionSpace, which replaces the factorized multi-discrete action parameterization with action-combination outputs; and SERPPO-noAM, which removes legality-aware action masking. In addition, the framework-level training curves also

Table 1

Compared baselines and internal ablated variants used in the experiments.

Category	Description	Alias
External baseline	Aggressive scripted controller based on a behavior tree	Aggressive BT
External baseline	Defensive scripted controller based on a behavior tree	Defensive BT
External baseline	PPO-based multi-agent reinforcement learning baseline	MAPPO
Internal ablation	Remove pseudo-termination-aware value estimation	SERPPO-noPD
Internal ablation	Replace factorized multi-head actions with action-combination outputs	SERPPO-1DActionSpace
Internal ablation	Remove legality-aware action masking	SERPPO-noAM

Table 2

Win rates against diverse opponent policies.

Opponent policy	HFO-SERPPO policy	Aggressive behavior-tree policy	MAPPO policy
Aggressive behavior-tree policy	0.82	0.51	0.69
Defensive behavior-tree policy	0.91	0.81	0.83
Stationary policy	0.97	0.95	0.90
Random-action policy	0.94	0.90	0.89
Randomly initialized neural policy	0.93	0.87	0.91
MAPPO policy	0.81	0.27	0.51
HFO-SERPPO policy	0.49	0.18	0.19

include SERPPO-noBO for completeness as an observation-related supplementary variant without obstacle information, while its detailed effect is deferred to the appendix-level supplementary analysis.

For the temporal extension, we additionally compare several variants of the actor-critic design. SERPPO denotes the non-recurrent base framework without temporal extension. SERPPO-LSTM replaces the feedforward actor with a history-aware LSTM actor, whereas SERPPO-GRU uses a GRU-based recurrent actor. HO-SERPPO denotes the variant with the history-aware actor but without future-action observation in the critic, and HFO-SERPPO denotes the full model with both the history-aware actor and the future-observing critic.

The overall training process follows the distributed rollout and PPO-style optimization pipeline described in Section 4.4 and Section 4.5. In the experiments, distributed rollout is treated as a supporting implementation choice for improving data-collection throughput rather than as an independent algorithmic contribution. We mainly report win rate and episodic return as the primary performance indicators. In addition, for pseudo-termination-aware value estimation, we inspect the critic loss to analyze value-fitting quality. For temporal modeling, we focus on convergence speed, training stability, and the attainable performance upper bound. For robustness evaluation, we test the learned policy under changes in opponent policy, team size, and environment layout. The compared baselines and internal ablated variants used throughout this section are summarized in Table 1.

5.2. Overall comparison results

We first compare the final HFO-SERPPO policy against diverse opponent policies to assess the overall effectiveness of the complete framework. The corresponding win rates are reported in Table 2. Unlike a narrowly defined single-opponent test, this evaluation covers both handcrafted behavior-tree opponents and learning-based opponents, and is therefore more suitable for examining whether the learned policy generalizes beyond a single training configuration.

Several observations can be drawn from Table 2. First, HFO-SERPPO achieves strong results against both aggressive and defensive behavior-tree opponents, reaching win rates of 0.82 and 0.91, respectively. This indicates that the learned policy is not merely fitted to a single scripted adversary, but is capable of adapting to substantially different opponent styles. Second, compared with the aggressive behavior-tree policy and the MAPPO policy, HFO-SERPPO also performs more strongly against MAPPO and the other simple baselines, which suggests that the complete framework provides a competitive end-to-end solution for this adversarial control setting.

At the same time, this overall comparison should be interpreted as evidence for the effectiveness of the complete

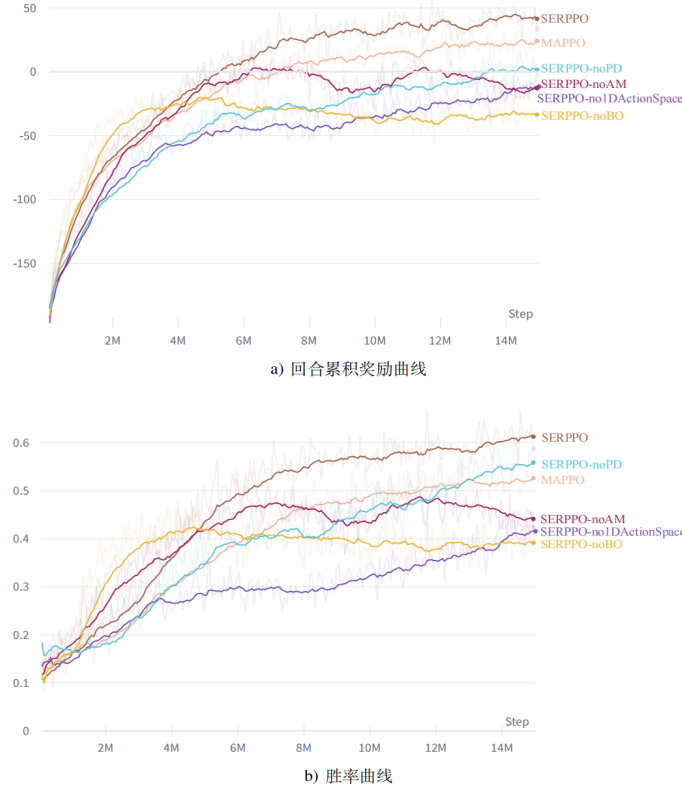


图 3-9 SERPPO 算法消融实验

Fig. 7: Framework-level ablation under mixed behavior-tree opponents. The curves compare the complete SERPPO framework with several internal ablations, while SERPPO-noBO is included as an observation-related supplementary variant.

framework rather than as a direct one-to-one validation of any single component. The role of the shared-and-exclusive reward formulation, the optimization-oriented modifications, and the temporal extension is further clarified in the following ablation studies.

5.3. Ablation studies

5.3.1. Framework-level ablation

To understand the role of the main components in the proposed framework, we first conduct a framework-level ablation study under a mixed opponent setting, where the sparring policy is aggressive or defensive with equal probability and remains unchanged within each episode. The map with complete obstacles is used throughout this comparison. The corresponding episodic-return curves and training win-rate curves are shown in Fig. 7. For completeness, this figure also includes SERPPO-noBO as an observation-related supplementary variant without obstacle information, although obstacle-representation analysis is not treated as a central methodological focus in the present paper.

From Fig. 7, the complete SERPPO framework exhibits the strongest overall training behavior among the compared variants. In both episodic return and training win rate, the full model converges to a better level than MAPPO, SERPPO-noPD, SERPPO-noAM, and SERPPO-1DActionSpace. This result suggests that the gain does not come from a single isolated modification, but from the joint effect of better reward semantics, more stable value estimation, and more suitable policy parameterization.

The comparison with MAPPO indicates that the complete SERPPO framework provides a stronger optimization target and better overall training behavior than a strong PPO-based MARL baseline in this task. The internal ablations further show that removing pseudo-termination weakens value estimation, removing action masking increases ineffective exploration, and reverting to action-combination outputs makes structured decision modeling less efficient. Detailed observation-related comparisons are deferred to Appendix A.

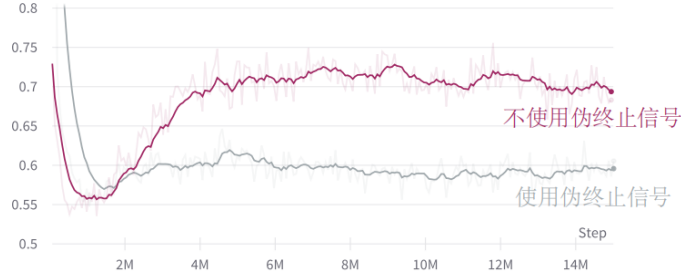


图 3-15 使用伪终止信号前后价值模型损失函数值曲线

Fig. 8: Critic-loss comparison with and without pseudo-termination-aware value estimation.

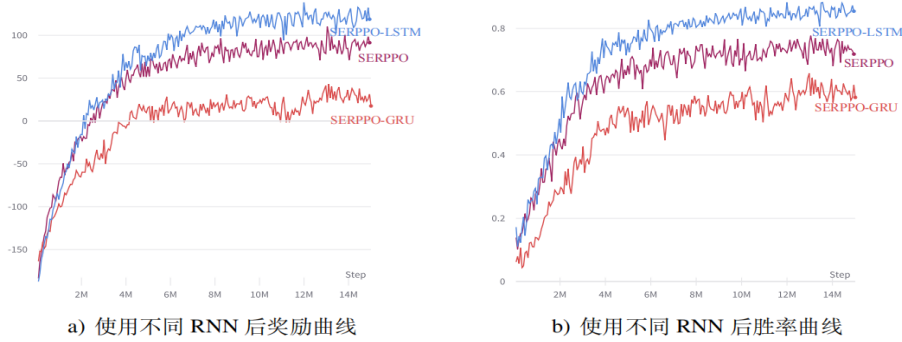


图 4-7 用 RNN 建模策略模型实验对比结果

Fig. 9: Comparison of different recurrent actor designs under partial observability.

5.3.2. Pseudo-termination-aware value estimation

The framework-level ablation in Fig. 7 already suggests that pseudo-termination-aware value estimation is beneficial. To further isolate its effect, we inspect the critic-loss curve before and after introducing pseudo-termination-aware value estimation. The comparison is shown in Fig. 8.

As shown in Fig. 8, the critic loss remains consistently lower after introducing pseudo-termination-aware value estimation. This indicates that the critic can fit the return target with smaller error when truncation and true termination are handled differently. Since the critic provides the bootstrap term for truncated returns and GAE computation, a lower critic loss here is consistent with improved value-target quality and reduced target ambiguity.

This result supports the design rationale in Section 4.2.1. By explicitly distinguishing genuine environmental termination from rollout truncation, the proposed formulation provides a more suitable target for value regression and thereby improves the reliability of advantage estimation used for policy optimization.

5.3.3. Temporal extension in HFO-SERPPO

We next examine the temporal extension introduced in Section 4.3, which consists of the history-aware actor and the future-observing critic. Since these two components address temporal uncertainty from different directions, they are discussed together here as the final stage that extends SERPPO into HFO-SERPPO. We first compare different recurrent designs for the actor. The results are shown in Fig. 9, where SERPPO-LSTM denotes the variant with an LSTM-based actor, SERPPO denotes the version without recurrent enhancement, and SERPPO-GRU adopts a GRU-based actor.

Fig. 9 shows that introducing an LSTM-based actor improves both training stability and the attainable performance upper bound, whereas replacing it with a GRU leads to more unstable training and inferior final performance in this setting. This comparison suggests that explicit history-aware modeling is beneficial for the task, and that the LSTM

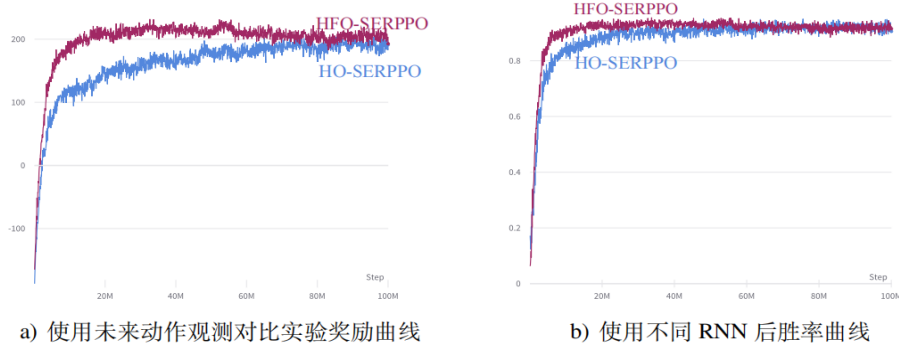


图 4-8 是否采用未来信息观测的实验对比结果

Fig. 10: Effect of future-action observation in the critic: HFO-SERPPO versus HO-SERPPO.

Table 3

Robustness to team-size variation.

Setting	1 RL-policy robot	2 RL-policy robots
1 aggressive enemy	0.73	1.00
2 aggressive enemies	0.15	0.82
1 defensive enemy	0.34	0.95
2 defensive enemies	0.25	0.91

design is empirically more suitable than the GRU design under the current setting. We then examine the effect of future observation on the critic. The comparison between HFO-SERPPO and HO-SERPPO is shown in Fig. 10, where HO-SERPPO includes the recurrent actor but does not incorporate future-action observation in the critic.

As shown in Fig. 10, introducing future observation significantly accelerates training. The reward and win-rate curves of HFO-SERPPO rise faster than those of HO-SERPPO, indicating that the future-observing critic provides a more informative training signal during optimization. The improvement in the final performance ceiling is comparatively moderate, suggesting that this component mainly improves optimization dynamics rather than directly increasing policy expressiveness. Taken together, Fig. 9 and Fig. 10 show that the history-aware actor and the future-observing critic contribute in complementary ways: the former strengthens execution-time inference under partial observability, whereas the latter reduces ambiguity in value learning during training.

5.4. Generalization and robustness analysis

5.4.1. Robustness to team-size variation

To further examine robustness beyond the main training setting, we evaluate the learned policy under different team-size configurations. Specifically, we consider one reinforcement-learning robot versus one advanced opponent, one reinforcement-learning robot versus two advanced opponents, and two reinforcement-learning robots versus one advanced opponent. To keep the observation space unchanged, two robots are still initialized on each side, but one robot can be set as already destroyed at the beginning of the episode. The results are summarized in Table 3.

The results in Table 3 show a clear gap between the single-robot and two-robot settings. When only one learned robot is available, performance decreases substantially, especially against two-enemy settings. By contrast, when two learned robots are deployed, the policy remains highly competitive across all tested cases. This suggests that the learned framework captures useful coordination behavior in the standard two-robot setting rather than merely improving isolated single-agent actions.

5.4.2. Robustness Under Environment Changes

We next evaluate the robustness of the learned policy under environment changes. In this simulator, the most direct way to modify the environment is to alter the obstacle layout. Therefore, we construct three new testing arenas by removing

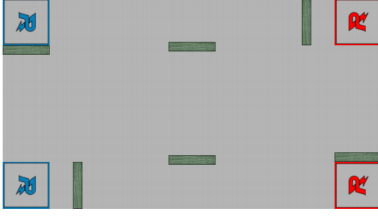


图 5-3 测试场地 1

(a) Arena 1.



图 5-4 测试场地 2

(b) Arena 2.



图 5-5 测试场地 3

(c) Arena 3.

Fig. 11: Modified testing arenas used for environment-robustness evaluation.

Table 4

Win rates under environment changes.

Arena	Defensive sparring policy	Aggressive sparring policy
Original arena	0.91	0.82
Arena 1	0.78	0.88
Arena 2	0.93	0.79
Arena 3	0.92	0.78

part or all of the obstacles from the original map. The three modified arenas are shown in Fig. 11. The corresponding win rates are reported in Table 4.

Table 4 shows that environment changes do affect policy performance, but the degradation is moderate rather than catastrophic. Against the defensive sparring policy, the learned model still maintains strong results in most modified arenas, while against the aggressive sparring policy the win rate declines slightly in two of the three changed arenas. This suggests that the learned policy captures transferable combat behaviors rather than simply overfitting to a single fixed map. Additional qualitative combat cases are provided in Appendix B.

5.5. Discussion

The experiments collectively indicate that the performance gain of the proposed framework comes from several complementary factors rather than from a single modification. At the framework level, the comparison with strong baselines shows that the complete method is competitive in multi-agent adversarial control. At the component level, pseudo-termination-aware value estimation improves critic fitting, structured action modeling and masking improve optimization efficiency, and the temporal extension improves learning behavior under partial observability.

Among these components, the temporal extension appears to contribute mainly through better optimization dynamics rather than through a dramatic change in final representational capacity. The history-aware actor improves the policy’s ability to exploit sequential evidence under local observation, whereas the future-observing critic reduces ambiguity in value learning and accelerates convergence.

At the same time, several limitations remain. The experiments are conducted in a simplified 2D simulator rather than in a higher-fidelity physical platform, and the reported results focus on the task configurations considered in this work. In addition, although robustness is evaluated under multiple opponent, team-size, and environment changes, broader cross-seed and cross-platform validation would further strengthen the empirical conclusions.

6. Conclusions and future work

This paper proposed a unified reinforcement learning framework for multi-agent real-time adversarial control. The framework addresses three coupled challenges in this setting, namely mixed reward semantics, structured decision spaces, and temporal partial observability. To this end, we developed SERPPO based on a shared-and-exclusive reward formulation, further improved optimization stability through pseudo-termination-aware value estimation, structured multi-discrete action modeling, and legality-aware action masking, and finally extended the framework to HFO-SERPPO

by combining a future-observing critic with a history-aware recurrent actor. Experimental results on a RoboMaster-style multi-robot adversarial simulator showed that the proposed framework consistently outperforms strong scripted and learning-based baselines, including behavior-tree policies and MAPPO. The ablation and robustness analyses further indicate that the improvement comes from the complementary effects of better reward semantics, more reliable value estimation, more suitable policy parameterization, and asymmetric temporal modeling under partial observability. In future work, the proposed framework can be evaluated in higher-fidelity simulators and real robotic platforms to further examine transferability under sensing noise, communication delay, and actuation constraints. It is also worth extending the framework to larger-scale adversarial scenarios with more complex cooperative-competitive interactions.

Acknowledgements

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Appendix

A. Supplementary ablation and training curves

This appendix provides supplementary training curves and auxiliary comparisons that support the analyses reported in Section 5. These results are not treated as central methodological evidence in the main text, but they provide additional context for observation design, actor–critic parameterization, and reward-related training behavior.

We first report supplementary comparisons on different obstacle representations, including lidar point clouds, local obstacle maps, and the setting without obstacle observation. These results provide additional evidence that obstacle-aware observations improve policy learning, while the lidar-based design offers a favorable trade-off between effectiveness and training efficiency in the current simulator setting.

We first provide supplementary comparisons on different obstacle representations, including lidar point clouds, local obstacle maps, and the setting without obstacle observation. These results further support the design choice discussed in the main text and show that obstacle-aware representations significantly improve the learning process, while lidar point clouds offer a better balance between effectiveness and training efficiency.

We next show the comparison between independent actor–critic parameterization and partially shared actor–critic parameterization. These curves indicate that the independent actor–critic structure trains more efficiently and reaches a better optimization trajectory, which is why the subsequent SERPPO-based exploration is built on the independent design.

Finally, we provide an additional comparison between SERPPO and MAPPO from the perspective of reward-related training behavior. This result serves as a supplementary illustration of why the shared-and-exclusive reward formulation leads to a more effective optimization process than the conventional MAPPO-style training objective.

B. Supplementary qualitative cases

This appendix further presents representative qualitative combat cases of the final HFO-SERPPO policy. These examples complement the quantitative results in the main text by showing typical policy behaviors against strong scripted opponents.

The first case corresponds to the defensive behavior-tree opponent, and the second corresponds to the aggressive

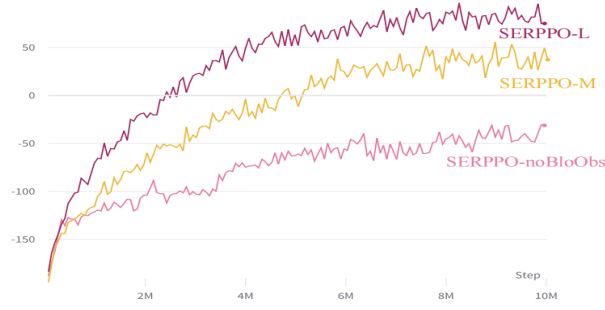


图 3-10 使用不同障碍物观测后奖励曲线

(a)

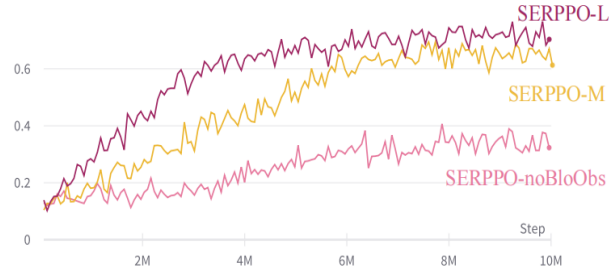


图 3-11 使用不同障碍物观测后胜率曲线

(b)

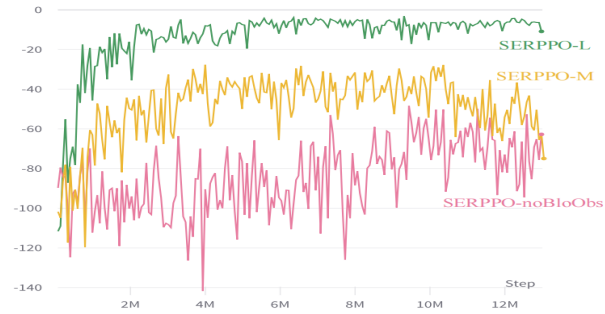


图 3-12 使用不同障碍物观测后有关障碍物的奖励曲线

(c)

Fig. A1: Supplementary comparison under different obstacle representations. (a) Episodic-return curves. (b) Training win-rate curves. (c) Obstacle-related reward curves.

behavior-tree opponent. Together, they illustrate the learned policy's coordinated movement, obstacle-aware navigation, and stable combat behavior under different engagement styles.

The first case corresponds to the defensive behavior-tree opponent, where the learned policy exhibits coordinated movement, obstacle-aware navigation, and stable attacking behavior. The second case corresponds to the aggressive behavior-tree opponent, where the engagement becomes more dynamic and better reflects evasive maneuvers, close-range confrontation, and the use of obstacle structures during combat.

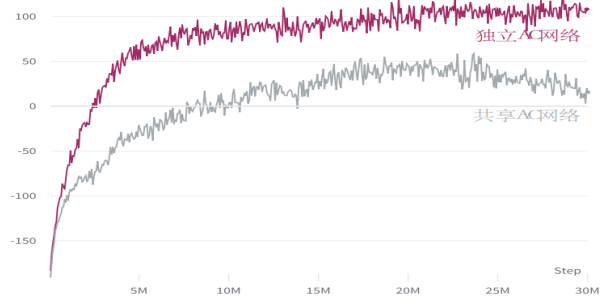


图 3-13 独立 AC 网络与共享 AC 网络奖励曲线

(a)

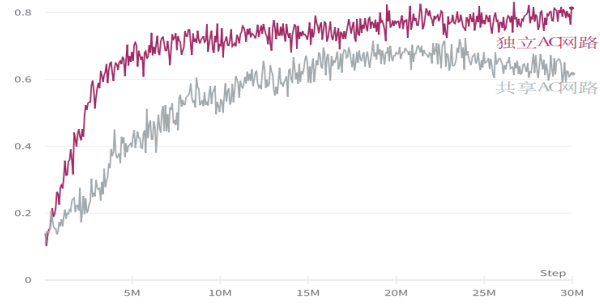


图 3-14 独立 AC 网络与共享 AC 网络胜率曲线

(b)

Fig. A2: Supplementary comparison between independent and partially shared actor–critic models. (a) Reward curves of independent and partially shared actor–critic models. (b) Win-rate curves of independent and partially shared actor–critic models.

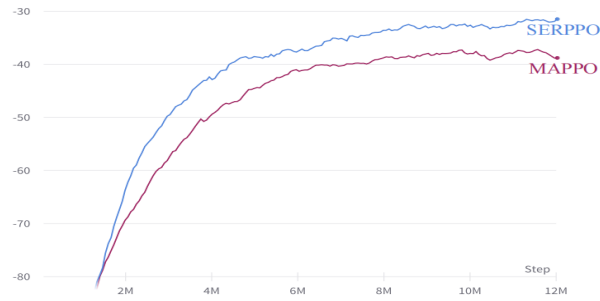


图 3-17 SERPPO 与 MAPPO 独享奖励曲线

Fig. A3: Supplementary comparison between SERPPO and MAPPO on reward-related training behavior.

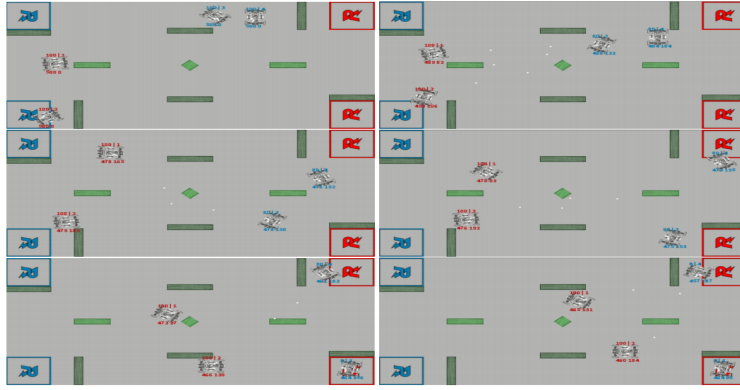


图 5-1 与防守型行为树对手对抗效果

(a)

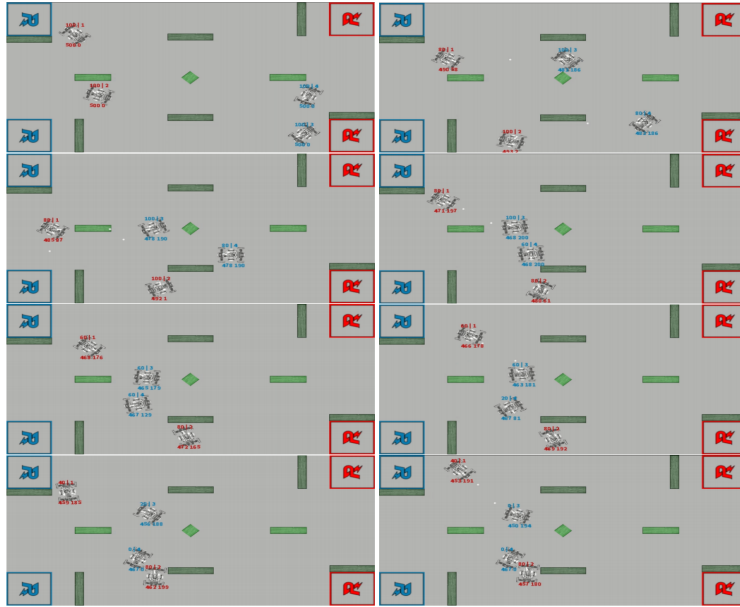


图 5-2 与进攻型行为树对手对抗效果

(b)

Fig. B1: Supplementary qualitative combat cases of the learned HFO-SERPPPO policy. (a) Representative case against the defensive behavior-tree opponent. (b) Representative case against the aggressive behavior-tree opponent.